

EDC BLOCK-003 (Derivation Attempt v2):

Can R_ξ Serve as the Compactification Scale L ?

Derivation attempt / exploratory note; not a results paper

EDC Research Program

2026-02-02

Status: *WORKING / OPEN*

1 Executive Summary

Question tested: Can the EDC weak-scale length $R_\xi = \hbar c/M_Z$ (or membrane thickness δ) serve as the compactification scale L in the brane-world reduction $G_N = \kappa_5^2/(6\pi L)$?

Outcome: INCONCLUSIVE.

Setting $L = R_\xi$ is dimensionally consistent and yields:

$$G_N = \frac{\kappa_5^2}{6\pi R_\xi}. \quad (1)$$

However, this merely trades one unknown (L) for another (κ_5^2). Without an EDC-internal expression for κ_5^2 (the 5D gravitational coupling), the derivation cannot close.

Therefore BLOCK-003 remains OPEN. The missing link is: $\kappa_5^2 = f(\sigma, R_\xi, \dots)$ from EDC geometry.

2 Recap: Why Non-Compact Fails

From derivation_v1 (Section 4), a non-compact extra dimension produces a 5D Green's function that yields:

$$\Phi(r) \sim \frac{\kappa_5^2 M}{r^2} \quad (\text{non-compact}), \quad (2)$$

which is *not* the observed Newtonian $1/r$ potential.

Key point [Dc]: The 5D Laplacian $\nabla_3^2 + \partial_\xi^2$ in Fourier space gives a propagator $\sim 1/k$ rather than $\sim 1/k^2$, leading to $1/r^2$ instead of $1/r$.

To recover 4D gravity, one needs either:

- A **compact** extra dimension of size L , or
- A **warped** geometry that localizes the graviton zero mode.

This document focuses on the compact case with L sourced from EDC.

3 Compact Recovery: The Standard Result

With a compact extra dimension $\xi \in [0, L]$ (orbifold or periodic), the zero-mode of the graviton dominates at distances $r \gg L$, and the effective 4D Newton constant is [Dc]:

$$G_N = \frac{\kappa_5^2}{6\pi L} = \frac{G_5}{L}, \quad (3)$$

where $\kappa_5^2 = 8\pi G_5$ and G_5 is the 5D Newton constant.

Dimensional Analysis

Dimensions (SI):

$$[G_N] = \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}, \quad (4)$$

$$[G_5] = \frac{\text{m}^4}{\text{kg} \cdot \text{s}^2}, \quad (5)$$

$$[\kappa_5^2] = [G_5] = \frac{\text{m}^4}{\text{kg} \cdot \text{s}^2}, \quad (6)$$

$$[L] = \text{m}. \quad (7)$$

Check: $[G_N] = [\kappa_5^2]/[L] = \text{m}^4/(\text{kg} \cdot \text{s}^2)/\text{m} = \text{m}^3/(\text{kg} \cdot \text{s}^2)$. ✓

The challenge: Equation (3) requires specifying *both* κ_5^2 and L independently. In standard Kaluza–Klein, L is a free parameter. For EDC to make G_N a *prediction*, both must come from EDC-internal physics.

4 Attempt A: $L := R_\xi$ (Primary)

Definition [BL]. The weak-scale length is:

$$R_\xi \equiv \frac{\hbar c}{M_Z c^2} \approx 2.16 \times 10^{-18} \text{ m}, \quad (8)$$

where $M_Z \approx 91.2 \text{ GeV}/c^2$ is the Z-boson mass.

Postulate [P]. Set $L = R_\xi$ as the compactification scale.

Consequence. Substituting into (3):

$$G_N = \frac{\kappa_5^2}{6\pi R_\xi}. \quad (9)$$

What this requires [OPEN]. For (9) to be predictive, we need κ_5^2 expressed in terms of EDC parameters. The most natural candidate would be the brane tension σ , via a relation like:

$$\kappa_5^2 \stackrel{?}{=} f(\sigma, R_\xi, \dots). \quad (10)$$

No such relation is derived in EDC. Therefore, (9) does not close BLOCK-003; it merely relocates the unknown.

Numerical check (illustrative only). If we *invert* to find the required κ_5^2 :

$$\kappa_5^2 = 6\pi R_\xi G_N \approx 6\pi \times (2.16 \times 10^{-18}) \times (6.67 \times 10^{-11}) \approx 2.7 \times 10^{-27} \text{ m}^4/(\text{kg} \cdot \text{s}^2). \quad (11)$$

This is a valid number, but it is *derived from* G_N^{obs} , violating anti-circularity.

Attempt A: INCONCLUSIVE.

Setting $L = R_\xi$ is dimensionally consistent but does not close BLOCK-003 because κ_5^2 remains unspecified by EDC. Inverting to find κ_5^2 from G_N^{obs} would be circular.

5 Attempt B: $L := \delta$ (Secondary)

Definition [P]/[I]. Let δ denote the membrane thickness in EDC. In some EDC contexts, δ is identified with a UV cutoff or junction width. Its precise value is **not uniquely fixed** in the current EDC literature.

Postulate [P]. Set $L = \delta$.

Consequence.

$$G_N = \frac{\kappa_5^2}{6\pi\delta}. \quad (12)$$

Status [OPEN]. This attempt has two unknowns: κ_5^2 and δ . Neither is derived from first principles in EDC. The situation is worse than Attempt A.

Attempt B: INCONCLUSIVE (worse).

Two unknowns (κ_5^2, δ) instead of one. No improvement toward closing BLOCK-003.

6 Warped Alternative (RS Scaling)

Alternative: Warped bulk geometry.

In Randall–Sundrum-type models, a warp factor $A(\xi)$ localizes the graviton zero mode near the brane. The effective 4D Newton constant takes the form:

$$G_N \sim \frac{\kappa_5^4 \sigma}{(\text{model-dependent factor})}. \quad (13)$$

This involves κ_5^4 (not κ_5^2) and the brane tension σ .

Caution [P]:

- Requires specifying $A(\xi)$ and solving the warped junction conditions.
- The “model-dependent factor” depends on AdS curvature, brane positions, etc.
- Not derived from EDC geometry here.

This remains a potential avenue but is **not pursued** in this note.

7 EDC Scale Candidates: Input Ledger

The following EDC-internal scales could, in principle, serve as L :

Input	Tag	Source	Used where	Circ. risk
R_ξ $\hbar c/M_Z$	[BL]	Z-boson mass (SM)	Attempt A	Low
δ (mem-brane)	[P]	EDC postulate	Attempt B	Medium
r_e (electron radius)	[BL]	EM sector	Not used here	High
$\sigma^{-1/3}$ (ten-sion scale)	[I]	Brane tension	Could try	Medium
$\kappa_5^{2/3}$	[OPEN]	5D coupling	Circular if in-verted	High

Key observation. All candidates either:

- Require κ_5^2 to complete the G_N formula, or
- Are already used in other EDC sectors (circularity risk).

8 Dependency Schematic

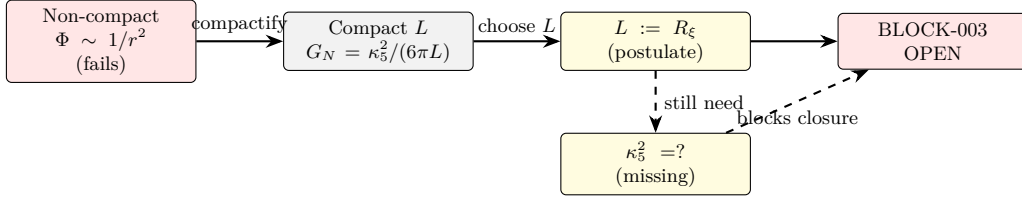


Figure 1: Derivation attempt schematic (program overview; not to scale; not a result). Setting $L = R_\xi$ is consistent but does not close BLOCK-003 without an independent source for κ_5^2 .

9 Outcome

Outcome: INCONCLUSIVE.

Attempt A ($L = R_\xi$):

- Dimensionally consistent.
- Produces $G_N = \kappa_5^2/(6\pi R_\xi)$.
- Does *not* close BLOCK-003 because κ_5^2 is unspecified.

Attempt B ($L = \delta$):

- Two unknowns (κ_5^2, δ); worse than Attempt A.

Missing link: An EDC-internal derivation of κ_5^2 in terms of $(\sigma, R_\xi, \dots)$.

Thus BLOCK-003 remains OPEN. The derivation is narrowed to: “Provide $\kappa_5^2 = f(\sigma, \dots)$ from EDC geometry.”

10 Conclusion

1. Setting $L = R_\xi$ (weak-scale length) is a reasonable postulate for EDC compactification.
2. This yields $G_N = \kappa_5^2/(6\pi R_\xi)$, which is dimensionally correct.
3. However, without κ_5^2 from EDC-internal physics, no prediction is made.

4. The warped (RS-type) alternative offers $G_N \sim \kappa_5^4 \sigma$, but requires specifying $A(\xi)$.
5. **BLOCK-003 remains OPEN.** The missing element is $\kappa_5^2 = f(\sigma, R_\xi, \dots)$.

Next step (out of scope for this note). Investigate whether EDC brane geometry or topological constraints can fix κ_5^2 . This would require analyzing the 5D bulk dynamics and their coupling to the brane tension.